



PROBLEM SPACE

SOCIAL ROBOT CO-DESIGN CANVASES

What problem are you solving?

ED nurses must rapidly assign ESI levels amid high patient volume and interruptions; the AI dashboard supports (never replaces) triage judgement.

USER

Group(s)	Characteristics	Needs
Who are the users? Are there supporting users? For example: students and teachers. ED Triage Nurses; Nurse-in-Charge (Sister P. Alleyne)	What are the users like? Clinically trained; work fast-paced, high-pressure shifts; frequently interrupted	What do the users need? Fast patient info; clear AI recommendations; ability to override; minimal extra documentation
primary users		
secondary users		
Emergency physicians; Observation Unit clinicians; hospital administrators; patients (indirect)	Make treatment and operational decisions; depend on accurate prioritisation	Explainable AI outputs; reliable triage data; audit trail for decisions
Goal(s) What do the primary and secondary users want to accomplish?		
Assess quickly; accurate ESI; reduce delays		Improve safety; reduce workload; consistent triage
primary users		
secondary users		
Reliable triage decisions; monitor patient flow		Improve ED efficiency; reduce adverse events; support QI initiatives
SHORT-TERM		LONG-TERM

ROBOT

Task(s)			
What task does the robot perform?			
Analyses demographics, symptoms, vitals and EHR data to predict an ESI level with a confidence score; nurse accepts or overrides.		Learns from outcomes to refine predictions over time; supports proactive risk flagging across the ED.	
SHORT-TERM		LONG-TERM	
Advantages			
What advantages does using a robot bring (compared to a computer or human)?			
Simple, understandable language; supports (not replaces) clinician communication	Increases confidence during busy shifts; reduces decision stress	Surfaces relevant patient info; prioritises high-risk cases	Rapid risk prediction; consistent ESI recommendations
Social skills	User's emotional response	Personalization	Precise tasks
Receives vital signs from connected patient monitors	Accessible from ED workstations and tablets	Not applicable	Integrates with Mercer EHR; connects to monitors; stores audit logs
Data collection with sensors	Mobility	Environment manipulation	Connection to systems

